

HARSH AGRAWAL

<designing proteins>

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</> Machine Learning, Protein Design, Computational Biology.

PUBLICATIONS

An Energy Landscape Approach to Miniaturizing Enzymes using Protein Language Model Embeddings

bioRxiv: doi:10.64898/2026.03.04.709378

March 2026

J. Lala, H. Agrawal, F. Dong, J. Wells, S. Angioletti-Uberti.

- Developed an AI-guided framework to miniaturize enzymes by sampling a protein language model (PLM) embedding energy landscape via Monte Carlo optimization (BAGEL), identifying maximally compact sequences that preserve catalytic site geometry — without requiring any structural information.
- Validated designs across four enzymes (PETase, subtilisin Carlsberg, Taq DNA polymerase, VioA) using a two-stage pipeline of deep-learning structure predictors (ESMFold, Chai-1, Boltz-2) and molecular dynamics simulations; top candidates achieved active-site RMSD < 2 Å and RMSF < 1.0 Å relative to wild-type. Open-sourced all generated sequences to accelerate experimental validation. *Experimental validation underway.*

EXPERIENCE

Encode: AI for Science Fellow

2026 - Present

Encode by Pillar VC

- Selected for Encode, a one-year AI-for-Science fellowship placing AI researchers in leading UK laboratories, built by Pillar VC and powered by ARIA and DSIT.
- Working on AI-driven design of improved gene delivery systems, in collaboration with Prof. Dame Molly Stevens (FRS, FREng; John Black Professor of Bionanoscience, University of Oxford).

Research Scientist

Feb 2025 - Present

Softnanolab, Imperial College London

- Currently developing a Protein Language Model designed natively for protein complexes. This work is supported by a 10K B200 GPU-hour Isambard Grant (Isambard AI) and a 16K A100 GPU-hour NVIDIA Grant, and is conducted under the supervision of Dr. Stefano Angioletti Uberti.
- Previously led the development of *ecstasy*, an open-source, comprehensive benchmarking framework for multimeric structure prediction methods (AF3, Boltz2, ESMFold, *et al.*), designed to assess robustness to chain-order permutations and consistency of predicted multimer assemblies.

Research Scientist

May 2024 - Jan 2025

Lab. of Medical Sciences, MRC Imperial College London

- Engineered novel, nanomolar-affinity binders for *C. elegans* transmembrane receptors by repurposing large protein toxins and designing new target-specific affinities. (*In collaboration with Dr. Karen Sarkisyan and Andre Brown.*)
- Designed the end-to-end computational pipeline for protein design and screening: evaluated binding interfaces using PPI databases, generated novel proteins (backbones and sequences) with state-of-the-art diffusion models (e.g., RFDiffusion), performed Rosetta relaxation, and conducted large-scale AlphaFold2 screening using metrics such as DockQ, iPTM, and iPAE.
- Built MLOps infrastructure from the ground up, scaling in-silico throughput by 50x (from ~10 to >500 designs/day) via distributed asynchronous inference across four heterogeneous compute clusters (>40 GPUs).
- Selected 10 designs for experimental validation by optimizing in-silico metrics (e.g., DockQ, iPTM) across a screening pool of ~10K candidates.

Lead Data Scientist

May 2021 - June 2022

Velora.io (formerly Paraswap.io)

EDUCATION

Masters in Engineering October 2022 - June 2026

Imperial College London

- Completed a Bachelor's + Master's (4 years) in Molecular Bioengineering. Fully sponsored by the RISE scholarship.
- Recipient of the Department Scholarship of 1000 GBP per year.
- Key modules include Advanced Synthetic Biology, Tissue Engineering, Biomaterials, Reinforcement Learning, Thermodynamics, etc.

PROJECTS

JESPR - Energy Based Modelling for Proteins

May 2023 - Dec 2023

- Contrastive pre-training of protein foundation models to learn sequence-structure representations enabling bi-directional mapping.
- Implemented sharding and activation checkpointing for multi-GPU training of 500M+ parameter models.
- Code: github.com/atom-51/jespr

Cancer-eX - Pancreatic Cancer Detection

Aug 2018 - Aug 2021

- Identified a salivary biomarker for early detection of pancreatic and GI cancers; designed and calibrated a competitive ELISA protocol and lateral flow assay.
- Built a risk-assessment tool using etiological risk factors weighted by relative-risk (RR) values from independent studies.

MineNow - The Gawler Challenge

Apr 2020 - Jul 2020

- Trained three ML models to predict ~300K novel mineralized sites in South Australia using geo-physical and geo-chemical datasets.
- Code: github.com/harshagrawal13/Gawler-Unearthed

ACHIEVEMENTS

- **Rise Global Winner 2021** July 2022
Selected as one of the 100 global winners for one of the largest high-school scholarships in the world by the Rhodes Trust and Schmidt Futures.
- **Bal Shakti Puraskar Recipient 2020** Jan 2020
Highest honor bestowed to an Indian below 18 by the President of India at the President's House. Awarded for my work on Cancer-eX.
- **Global Teen Leader 2020** Jan 2020
Selected as one of the 35 teens all around the world by Three Dot Dash to participate at the Virtual Just Peace Summit 2020. Was part of a youth delegation to speak at the World Economic Forum 2024 in Davos, Switzerland.
- **Regional Finalist, Google Science Fair** May 2019
Selected as one of the top 100 exhibits all around the world at the Google Science Fair.

- Built the public analytics pipeline and **dashboard** for Velora, the second-largest DEX aggregator, architecting a robust system indexing 33M+ transactions and \$140B+ in trading volume across 5 blockchains.
- Co-led the upgrade of the full data infrastructure from Heroku to AWS, deploying a scalable Redshift + MongoDB warehouse to support low-latency querying, reliable ingestion, and long-term storage of high-volume on-chain data.
- Developed internal dashboards powered by **Tenderly** simulations to model transaction outcomes and MEV behavior, directly informing decisions on routing optimisation, onboarding new DEXs, and smart contract improvements.